OS 3 比較題

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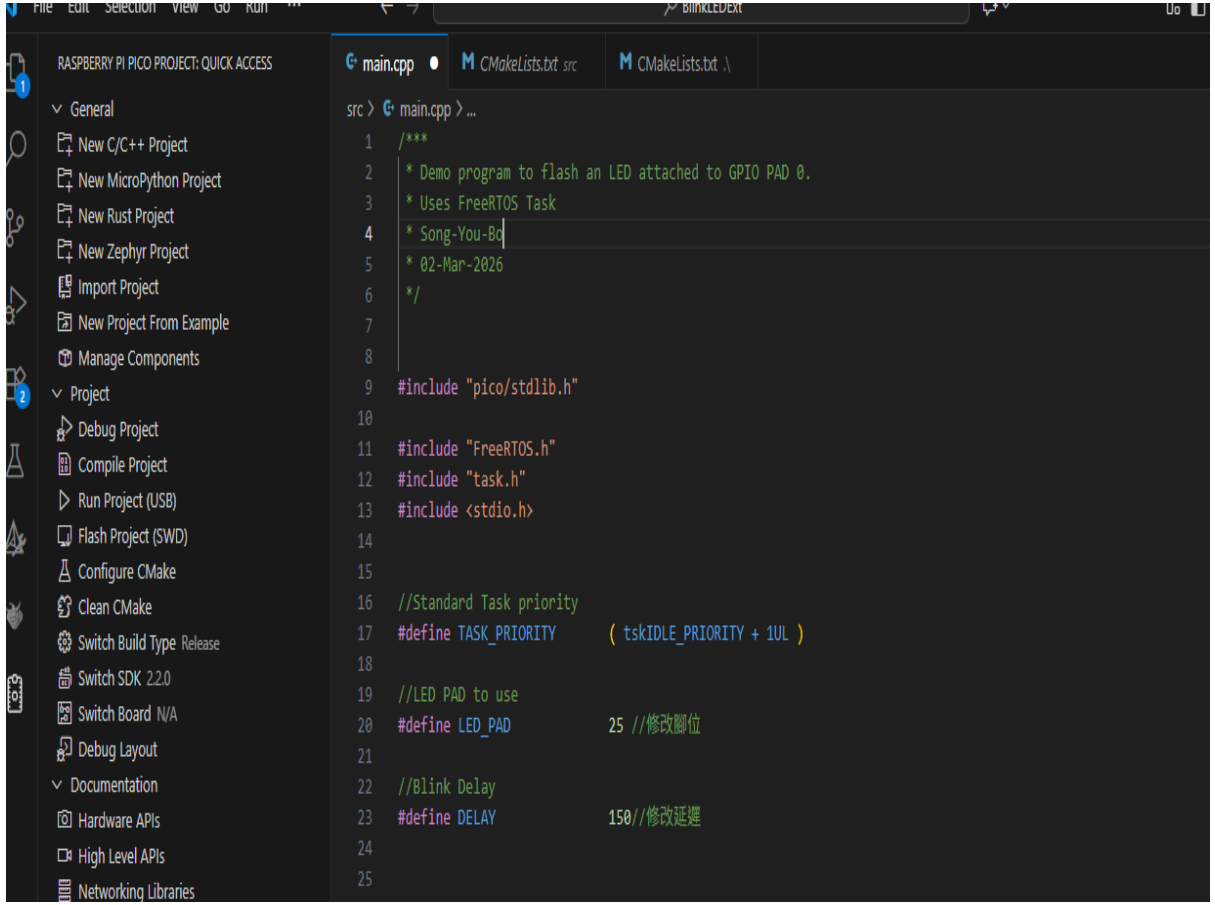
File Edit Selection View Go Run
BlinkLEDext
RASPBERRY PI PICO PROJECT: QUICK ACCESS
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New Rust Project
New Zephyr Project
Import Project
New Project From Example
Manage Components
Project
Debug Project
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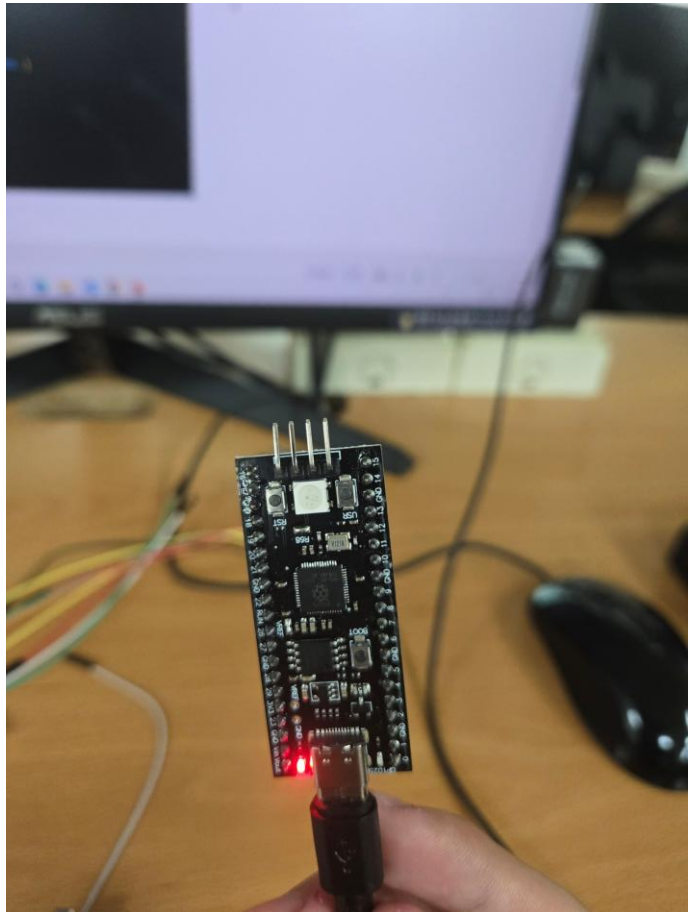
Run Project (USB)
Flash Project (SWD)
Configure CMake
Clean CMake
Switch Build Type Release
Switch SDK 2.2.0
Switch Board N/A
Debug Layout
Documentation
Hardware APIs
High Level APIs
Networking Libraries
main.cpp
CMakeLists.txt src
CMakeLists.txt .\
src > main.cpp > ...
/***
* Demo program to flash an LED attached to GPIO PAD 0.
* Uses FreeRTOS Task
* Song-You-Bo
* 02-Mar-2026
*/
#include "pico/stdlib.h"
#include "FreeRTOS.h"
#include "task.h"
#include <stdio.h>
//Standard Task priority
#define TASK_PRIORITY (tskIDLE_PRIORITY + 1UL)
//LED PAD to use
#define LED_PAD 25 //修改腳位
//Blink Delay
#define DELAY 150//修改延遲



main.cpp ● CMakeLists.txt ×

CMakeLists.txt

```cmake
# == DO NOT EDIT THE FOLLOWING LINES for the Raspberry Pi Pico VS Code Extension to work ==
if(WIN32)
    set(USERHOME $ENV{USERPROFILE})
else()
    set(USERHOME $ENV{HOME})
endif()
set(sdkVersion 2.2.0)
set(toolchainVersion 14_2_Rel1)
set(picotoolVersion 2.2.0)
set(picoVscode ${USERHOME}/.pico-sdk/cmake/pico-vscode.cmake)
if (EXISTS ${picoVscode})
    include(${picoVscode})
endif()
# ==============================================================================
set(PICO_BOARD pico CACHE STRING "Board type")

cmake_minimum_required(VERSION 3.12)

# Change your executable name to something creative!
set(NAME BlinkLEDExt) # <-- Name your project/executable here!

include(pico_sdk_import.cmake)

# Gooey boilerplate
project(${NAME} C CXX ASM)
set(CMAKE_C_STANDARD 11)
set(CMAKE_CXX_STANDARD 17)

# Initialize the SDK
pico_sdk_init()

#include Libraries
#SET(FREERTOS_KERNEL_PATH "${CMAKE_CURRENT_LIST_DIR}/../../lib/FreeRTOS-Kernel" CACHE STRING "Course Common Lib")
SET(FREERTOS_CONFIG_FILE_DIRECTORY "${CMAKE_CURRENT_LIST_DIR}/port/FreeRTOS-Kernel" CACHE STRING "Local Config")
include_directories("${FREERTOS_CONFIG_FILE_DIRECTORY}")
include(FreeRTOS_Kernel_import.cmake)
```

```cmake
#Add main source directory
add_subdirectory(src)

#Set up files for the release packages
install(CODE "execute_process(COMMAND $ENV{HOME}/bin/picoDeploy.sh ${CMAKE_CURRENT_BINARY_DIR}/src/${NAME}.elf)")

# Set up files for the release packages
install(FILES
    ${CMAKE_CURRENT_BINARY_DIR}/src/${NAME}.uf2
    DESTINATION ${CMAKE_CURRENT_BINARY_DIR}
)

set(CPACK_INCLUDE_TOPLEVEL_DIRECTORY OFF)
set(CPACK_GENERATOR "ZIP" "TGZ")
include(CPack)
```

README

比較項目	**Ch0 SDK (基礎版)**	**Ch2 BlinkLEDExt (模組版)**	**Ch4 BlinkExt (實戰版)**
作業系統 **(OS)**	無 (Bare Metal)	無 (Bare Metal)	**FreeRTOS**
程式架構	單一 while 迴圈	模組化設計 (Modular)	任務導向 **(Task-based)**
延遲方式	sleep_ms() (忙碌等待)	sleep_ms()	**vTaskDelay() (釋放CPU)**
多工能力	無 (一次只能做一件事)	無	有 **(可同時處理多項任務)**
系統複雜度	低	中	高
實務應用	教學、基礎硬體控制	小型專案、單一功能設備	工業應用、複雜系統